Bundle-Net research paper

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1 Introduction

A central challenge in modern neuroscience is understanding how brain activity drives behavior, ranging from simple sensory or motor tasks to more complex cognitive processes, such as interacting with others or making decisions. Existing research demonstrates that the incorporation of brain-behavior relationships [[12], [23], [19]] into joint modeling frameworks is a critical step toward achieving this goal.

While the number of available analytical approaches is large, recent advances have established artificial intelligence (AI), and deep learning in particular, as the highest-performing methods. These tools enable researchers to jointly model neural dynamics and behavior [[37],[8]], as well as perform dimensionality reduction, using techniques such as autoencoder architectures [[38], [39], [20]], contrastive learning frameworks [[28]] or newly introduced causal representation learning.

However, the "black box" nature of deep learning models makes it extremely difficult to trace the decision-making process and identify key features. Although excellent performance is important, it is no longer sufficient to assess a model solely on the basis of that metric. Mathis and Mathis (2026) [[18]] argue that deep learning models and their resulting latent spaces must also be assessed in terms of their trustworthiness and interpretability. This ensures that outcomes are consistent, robust to noise, identifiable, and both human-interpretable and mathematically explainable [[28], [9], [15]].

Expanding on this idea, this research aims to fill a significant gap in the literature by extending the existing analysis of the BunDLe-Net framework with post-hoc explainability methods. By applying this approach to calcium imaging data from C. elegans, this study directly addresses the growing need for deep learning models to be thoroughly evaluated based on their microscopic interpretability.

The rest of the paper is structured as follows. Section 2 reviews the relevant literature on neural manifold modeling, dimensionality reduction, and post-hoc Explainable AI (XAI) methods. Section 3 outlines the dataset and describes the methodology used to apply and evaluate the chosen XAI techniques. Section 4 presents the empirical results, including performance assessments of the XAI methods and a comparative analysis of results utilizing an interactive dashboard. Finally, Section 5 concludes the paper, summarizing the main findings and outlining potential directions for future research.

2 Literature Overview

Computational ethology is the science of understanding how to automate the measurement and analysis of animal behaviour, especially in relation to brain activity [[3]]. Scientists in this field have established that although studying single-neuron signals has yielded meaningful results, the shaping of motor programs corresponds to neural activity patterns across large neuronal populations. These populations consist of hundreds of sampled neurons that act in a coordinated manner and encode information dynamically [[4],[5],[17],[31]].

A particularly important contribution to this domain, and highly relevant to the present study, was made by Kato et al. [[10]]. In their research, calcium imaging data from C. elegans was collected and analyzed to demonstrate that the high-level organization of behavior is encoded in the brain by globally distributed, continuous, and low-dimensional dynamics. In other words, to model the behavior of C. elegans, we can use low-dimensional spaces that capture the coordinated activity of their brain - commonly referred to as neural manifolds [[5], [30]].

Based on these findings, it is evident that not all neuronal activity patterns are equally important, making it crucial to filter out biological noise. Therefore, successful brain activity modelling requires dimensionality reduction to capture only those features that are relevant in a behavioral context. As utilizing neural manifolds has proven to be a highly promising approach in achieving this, neuroscience incorporates a wide range of dimensionality reduction techniques, both linear and non-linear [[21]].

Linear dimensionality reduction methods, such as Principal Component Analysis (PCA) [[22]], Factor Analysis (FA) [[32]], and Multidimensional Scaling (MDS) [[13]], are relatively simple, yet they remain widely employed in neuroscience research. As demonstrated by Ahrens et al. (2012) [[1]], PCA can be successfully used on calcium imaging data to visualize the activity of all identified brain sites by embedding it into a three-dimensional phase space. Although PCA effectively allowed them to summarize massive amounts of data into just three time series, researchers have pointed out that PCA suffers from fundamental limitations. Because it strictly maximizes total variance, PCA captures Poisson-like spiking noise alongside true neural signals, and it often fails to detect small, sparse activations of individual neurons, which was also mentioned as a biggest disadvantage of this method in a review done by Cunningham and Yu (2014) [[6]]. However, since this is a commonly used method to coarse-grain data, PCA can serve as a great first baseline model for more complex approaches [[14]].

More promising methods emerge from non-linear group, as neural data often lie on low-dimensional, non-linear manifolds embedded within high-dimensional spaces. While Cunningham and Yu (2014) [[6]] highlighted Isomap [[35]] and Locally Linear Embedding (LLE) [[26]] as two of the most prominent classical non-linear techniques, there have been significant methodological developments since their review.

As outlined by Mathis and Mathis (2026) [[18]], modern autoencoder-based techniques that aim to reconstruct input data from learned latent representations have overcome the limitations of classical linear and non-linear methods. The authors specifically emphasized Variational Autoencoders (VAEs) [30] and Latent Factor Analysis via Dynamical Systems (LFADS) [[20], [11]] and classified them as generative models or reconstruction-based approaches. By utilizing these deep learning architectures, researchers have successfully captured complex non-linear dynamics by compressing and subsequently reconstructing original neural signals. However, Mathis and Mathis also point out a fundamental flaw: because these models prioritize perfect data reconstruction, they often overlook features that are most crucial for behavior, inadvertently encoding biological noise into the latent space.

To tackle the persistent problem of biological noise, researchers have developed another class of approaches, referred as contrastive learning. This group, in contrast to autoencoder-based approaches, does not attempt to perfectly reconstruct the raw input data. Instead, they aim to discover relationships within the data by directly aligning neural activity with behavioural variables [[18]]. A state-of-the-art approach within the contrastive model's group is a method called CEBRA [[11]]. This neuronal manifold technique optimizes the encoding of data by maximizing the similarity between related samples and minimizing the similarity between unrelated samples. Furthermore, to incorporate both behavioural and time dynamics, a variant known as CEBRA-hybrid can be utilized [[29]].

Within the group of the most novel approaches in neural manifold learning, alongside generative and contrastive learning, we can also find an emerging subgroup known as causal representation learning, or Markovian state representations. This approach is currently best represented by the Behaviour and Dynamics Learning Network (BunDLe-Net) [[14]]. Building on the idea introduced by Krakauer et al. (2017) [12] that neuroscience needs behaviour, the authors of BunDLe-Net designed a tool that learns low-dimensional Markovian embeddings of neural dynamics. By doing so, it successfully preserves exclusively the aspects relevant to a specific behavioural context, effectively de-noising the data.

In the work presented by Kumar et al. (2023) [[14]], their framework (BunDLe-Net) significantly outperformed other tested methods, including state-of-the-art CEBRA, across various neuronal datasets ranging from C. elegans and rats to primates. However, the researchers not only achieved high-performance metrics, but they also effectively addressed the trustworthiness and interpretability of the model.

Mathis and Mathis (2026) [18] argue that the trustworthiness of a model should be evaluated by three main metrics: consistency, robustness, and identifiability. The authors of BunDLe-Net tested the consistency of the created latent spaces by comparing them across different initializations and data from five different worms. Although different initializations naturally result in visually different embeddings, each reconstruction displayed a highly consistent branching structure in the trajectories, completely independent of the initialization. Furthermore, the authors ensured that the framework is highly robust to noise and dropout. First, to prevent overfitting, they injected Gaussian white noise into the mapping layer during training, forcing the model to generalize rather than learn noise from the training data. Second, the model intrinsically acts as a de-noising tool by eliminating neuronal dynamics that are irrelevant to decoding behaviour. Finally, the authors addressed the identifiability of the model (its capacity to uncover the true underlying data). In the discussion section of the article, they attribute the ability to identify the true, intrinsic structure of the neuronal manifold to the time-delayed embeddings of the neural dynamics used for learning the latent space. This approach, backed by Takens' theorem [34], guarantees that BunDLe-Net's reconstructions will be topologically consistent regardless of the specific subset of recorded neurons.

The authors of BunDLe-Net claim that the embeddings generated by their model are "useful and interpretable". This assertion is supported by both the theoretical principles underlying the framework and the visual clarity of the resulting manifolds. Theoretically, the model is designed so that the latent embedding encapsulates all necessary dynamic information from the raw neuronal data, allowing it to function as a distinct and meaningful dynamic process on its own. Furthermore, the embeddings produced by BunDLe-Net naturally organize into well-separated "bundles". This lack of artificial trajectory intersections makes it easy for researchers to distinguish behavioural states directly from the visualized manifold.

However, the form of interpretability defined by the authors of BunDLe-Net does not fully satisfy the broader need for transparent deep learning models in neuroscience, as emphasized by Mathis and Mathis (2026) [18]. While the macro-level dynamics and the visualization of the neural manifold are highly interpretable, the underlying model itself remains a "black box". To fully evaluate BunDLe-Net against these rigorous transparency standards and understand its internal decision-making, the explainability of the model must be systematically tested using post-hoc Explainable AI (XAI) methods.

Explainable Artificial Intelligence, also referred to as XAI, is a group of methods and frameworks designed to help understand the decision-making process behind complex machine learning algorithms [24]. As argued by Alharthi et al. (2025) [2], the integration of XAI into healthcare research is a much-needed development, yet it remains frequently overlooked. The growing complexity of AI models makes it increasingly difficult for them to be transparent, interpretable, and therefore trustworthy; thus, the authors view XAI methods as a necessity to overcome these problems.

To address these transparency issues, the majority of research connecting neuroscience with XAI currently relies on established, foundational methods, such as Integrated Gradients (IG) [[33]] or Shapley Additive exPlanations (SHAP) [[16]]. SHAP is a model-agnostic approach that provides both local and global explanations; however, it can be computationally expensive when applied to large models and datasets [[2]]. Zhang et al. (2026) [[41]] demonstrated that the SHAP method can be exceptionally useful for identifying the most critical features in time-series data. In their work, they employed an XGBoost model to predict whether patients with migraine had comorbid depressive symptoms. Recognizing the clinical limitations generated by the model's inherent "black-box" nature, they integrated SHAP to quantify feature contributions. The authors highlighted that using SHAP significantly increased clinical transparency by providing directional insights - specifically, by demonstrating exactly how higher or lower fractional occupancy in certain brain states influenced the model's predictions toward a specific diagnosis.

Another compelling application of these foundational XAI methods in neuroscience was presented by Rahman et al. (2022) [[25]]. To maintain stable, ecologically valid interpretations while achieving high performance, the authors used Integrated Gradients (IG) and SmoothGrad on IG (SGIG) within their deep learning framework. Gradient methods prove to yield consistent measurements and perform exceptionally well with neural networks, although they can be less intuitive compared to SHAP due to the requirement of selecting a reference baseline [2]. The authors applied these interpretability techniques to resting-state functional magnetic resonance imaging (fMRI) data from patients with autism spectrum disorder, schizophrenia, and Alzheimer's disease (multivariate time-series data). By introspecting the model, they successfully extracted sparse spatio-temporal signatures that were highly indicative of these disorders. Proving through a "Retain And Retrain" (RAR) validation method, they showed that just the top 5% of the salient data identified by XAI was highly predictive of the diseases. The researchers emphasized that IG and SGIG were specifically chosen because they proved to be stable, generalizable, and robust to noise across all datasets. Ultimately, they highlighted that applying these methods resolves the vanishing saliency problem typical of time-series models, facilitating the discovery of actionable, causal knowledge about brain dysfunction directly from raw data.

Despite these significant achievements with established post-hoc methods, there is still limited XAI research within the joint-modelling subgroup. A highly notable study in this domain is the work by Schneider et al. (2025) [29]. The researchers addressed the "black-box" nature of their own contrastive learning framework, CEBRA, by creating a new variant called xCEBRA (explainable CEBRA). By applying a novel Inverted Neuron Gradient attribution method, they successfully generated time-series attribution maps capable of identifying the specific causal factors driving the model's predictions, outperforming standard Shapley value-based approaches.

Given the rapid expansion of both foundational and novel XAI tools, adopting a broader comparative approach has become essential, as highlighted by Šarčević and Frid (2026) [27] and Alharthi et al. (2025) [2]. These articles focused on evaluating various XAI methods applied to neuroscience and healthcare sensor data, explicitly outlining their respective strengths and weaknesses. Both studies concluded

that since different XAI algorithms rely on distinct assumptions and often yield divergent attribution patterns, it is crucial to employ multiple methods to ensure reliable interpretations. Furthermore, the authors emphasized that systematically evaluating the XAI results themselves is essential and should not be overlooked. To address this, Šarčević and Frid (2026) [27] implemented a multi-metric evaluation framework to test XAI stability across categories such as fold-wise stability, attribution compactness, and cross-method ranking agreement. However, as Alharthi et al. (2025) [2] highlighted, there remains an urgent need to establish a universally standardized evaluation and regulatory framework across all XAI methods, which is currently lacking in the field.

Considering the findings from the described studies, this research aims to fill a significant gap by systematically applying post-hoc XAI methods to the BunDLe-Net framework. To achieve this, the study will employ two well-tested but distinct approaches: Integrated Gradients (IG) and Shapley Additive exPlanations (SHAP). These methods were selected due to their extensive validation in existing literature and the availability of multiple ideas of how to properly evaluate their performance. Crucially, both IG and SHAP are highly compatible with BunDLe-Net's specific input format, which relies on flattened sliding time windows of neural activity. These attribution techniques can effectively map the relevance scores back to both the specific spatial features (individual neurons) and the precise temporal instances within those windows. Ultimately, this approach aims to make BunDLe-Net fully interpretable on a microscopic level, revealing exactly which individual neurons drive the model's decisions and at what specific moments in the time-series these activations occur.

3 Dataset and methodology

4 Empirical Results

The Explainable AI (XAI) attribution methods were evaluated across four configurations of the Bunddle-net model's hyperparameter $\gamma \in \{0.7, 0.8, 0.9, 0.95\}$. The γ parameter controls the relative weight of the loss functions, balancing the behavioral loss against the Markovian latent dynamics loss; higher values of γ place greater emphasis on reconstructing behavioral outputs. Using the interactive dashboard, we evaluated four distinct research hypotheses to assess model behavior and interpretability. For the analysis the results with γ equal to 0.9 were used, however additional analysis across diffrent gammas is included at the end of the section.

4.1 What neurons are most important for each behavioral state?

From a neurobiological standpoint, the underlying ground truth for this question is well-established: Kato et al. (2015) [10] demonstrated that distinct motor behaviors in *C. elegans* are driven by specific, coordinated subgroups of neurons (e.g., forward locomotion is primarily driven by AVB interneurons and RIB neurons), with high consistency across individual worms. However, whether a deep generative model like Bunddle-net faithfully mirrors this biological reality remains an open question. Evaluating this hypothesis directly addresses model alignment and trustworthiness, identified by Mathis and Mathis (2026) [18] as critical prerequisites for deploying joint brain–behavior AI models. Specifically, we test whether feature attribution profiles for a given behavior remain highly correlated across individual worms, which would point out that model is generalizing,

It is important to note that during the training of BunDLe-Net, a specific subset of neurons (AVAR, AVAL, SMDVR, SMDVL, SMDDR, SMDDL, RIBR, and RIBL) was deliberately excluded from the input data to prevent redundancy, given their direct role in behavioral classification. Consequently, while these neurons are biologically crucial for certain behaviors, they were absent from the input space, meaning the applied XAI methods could not assess or score their importance. In the Table I the excluded neurons are highlighted with a red color.

Behaviour	Kato et al. (2015)
Dorsal Turn	RMED
Forward	AVB, RIB (excluded), RMEV
No state	-
Reverse 1	AVA (excluded), AVE, RIM, AIB
Reverse 2	AVA (excluded), AVE, RIM, AIB
Sustained Reversal	AVA (excluded), AVE, RIM, AIB
Slowing	AIB, RIB (excluded)
Ventral Turn	SMDV (excluded), RIV

Table 1: Reference neurons for each behavior based on Kato et al. (2015).

4.1.1 Diminishing Saliency and the Low-Dimensional Neural Manifold

Before evaluating the consistency of the model across different worms and behaviors, it is crucial to understand how BunDLe-Net distributes its predictive focus across the neural population. Initial analysis of the Explainable AI (XAI) attributions reveals a distinct phenomenon of *diminishing saliency*. As demonstrated in Figure 1, the feature importance scores (both SHAP and Integrated Gradients) exhibit a sharp, heavy-tailed decay. Across all recordings, only a marginal fraction of the neural population, approximately 5% to 10%, carries meaningful predictive weight, while the vast majority of neurons form a long tail of near-zero importance.

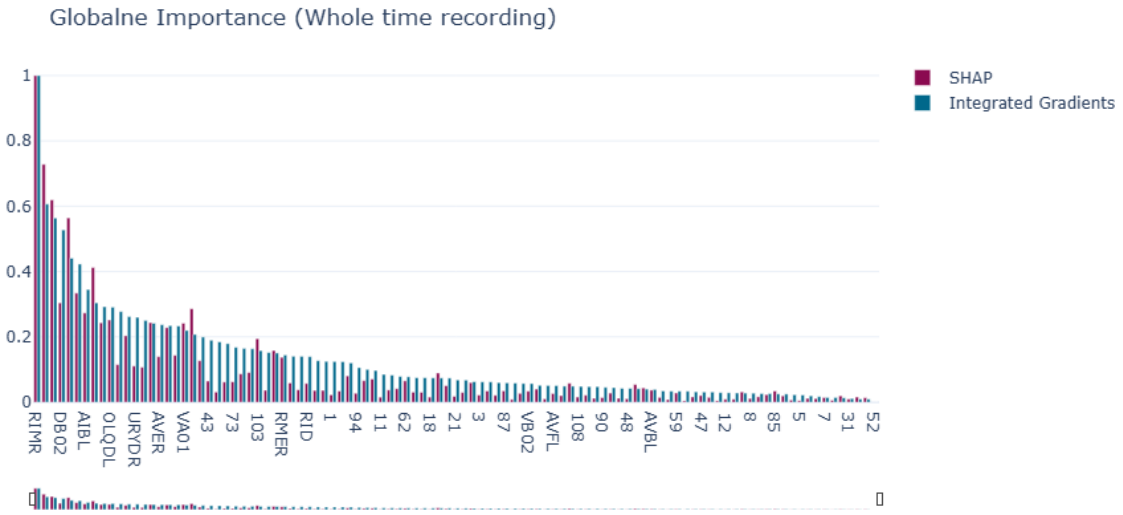


Figure 1: Distribution of SHAP and IG values per neuron for a single recording (Worm I), illustrating the exponential decay in feature importance.

To quantitatively validate that this sparse representation captures the true decision-making process of the model rather than being an artifact of the explainer, we conducted a Retain-And-Retrain (RAR) evaluation following the protocol introduced by Rahman et al. (2022) [25]. Strikingly, when restricting the input space to only the top 10% most salient neurons, the model maintained an accuracy of 95% or higher.

This computational sparsity aligns perfectly with fundamental principles of systems neuroscience. As highlighted by Cunningham and Yu (2014) [6], neural population activity typically resides on a low-dimensional manifold governed by a small number of latent collective drivers, rendering much of the high-dimensional single-neuron activity redundant. Furthermore, our findings corroborate the observations made by Kato et al. (2015) [10] in *C. elegans*. Their research demonstrated that although the activity of over 100 neurons can be recorded simultaneously, just three principal components are sufficient to explain over 65% of the variance, implying that only a small, core subset of neurons actively drives global behavioral states.

The fact that BunDLe-Net preserves its full predictive power when restricted to this minimal neural subset confirms that the model's latent representation successfully isolates the core computational manifold of the worm's nervous system. Consequently, this extreme concentration of feature importance provides a vital methodological foundation for our subsequent analyses. To accurately assess the stability and biological validity of the model, the following sections will focus specifically on the highly salient neural core using rank-aware and Top-k metrics.

4.1.2 Overall comparison

In tables below the results of XAI methods are presented. Each table corresponds to one behavior. Note that neurons idetified as most importnat by Kato et al. are highlighted in the bold blue color.

	Worm 1		**Worm 2**		**Worm 3**		**Worm 4**		**Worm 5**	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	OLQVL	OLQVL	AIBR	VB01	**RMED**	RMEV	**RMED**	53	AIBR	**RMED**
2	OLQDL	DB02	OLQDL	OLQDL	RIMR	**RMED**	DA01	DA01	AVER	RMEV
3	AIBR	**RMED**	OLQVR	RMEV	AVER	RIMR	AS01	OLQVL	**RMED**	AIBR
4	OLQVR	OLQDL	RIMR	AIBR	RIVL	VB01	SIBVL	AS01	24	24
5	RIMR	AIBR	ASKR	54	18	AVER	OLQVL	SIBVL	20	AVER
6	RIML	AVEL	RIVR	10	31	AVBR	AIBR	RIVL	VA01	20
7	AVEL	RIML	66	RIMR	AVEL	AIBR	53	RMER	22	RIMR
8	RIVR	OLQVR	54	RIFR	128	RIVL	SABVR	RIVR	126	49
9	99	RIMR	RMER	OLQVR	125	26	59	AVBR	ALA	18
10	AIBL	RIVR	7	RIVR	RMEV	RMER	RIVL	RIMR	AVFR	RIML

Table 2: Comparison of Top 10 rankings for **Dorsal turn** behavior

	Worm 1		**Worm 2**		**Worm 3**		**Worm 4**		**Worm 5**	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	DB02	RIMR	DB01	RIMR	VB02	VB02	61	RIML	127	127
2	RMED	RIML	RMER	DB01	RMED	RMED	RMED	ASKL	**RMEV**	VA01
3	**AVBR**	AIBR	SIBDR	AIBR	**RMEV**	RIML	VB02	RID	RMED	**RMEV**
4	URYDR	RMED	RIVR	RIML	82	RIMR	SIBVL	AIBR	VA01	RIMR
5	43	RIVR	AIBR	RMER	83	RIVL	RIS	RMED	54	RMED
6	109	AVEL	VB01	RIVR	76	AIBL	7	117	**AVBL**	AIBR
7	ASKL	OLQDL	66	54	RIVL	AIBR	URYDL	VB01	87	1
8	RID	**AVBR**	VB02	RIFR	SIADL	83	ALA	RIVL	15	**AVBL**
9	RIMR	AIBL	RMEL	VB02	VB01	**RMEV**	ASKL	SIBVL	129	RIVL
10	DB01	**RMEV**	VB03	VB01	**AVBR**	82	AVFL	RMEL	120	ASKR

Table 3: Comparison of Top 10 rankings for **Forward** behavior

	Worm 1		**Worm 2**		**Worm 3**		**Worm 4**		**Worm 5**	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	**RIMR**	**AIBR**	RIVR	RIVR	RMEV	RMED	RIVL	RMED	VA01	RMED
2	**AIBR**	**AVEL**	**AIBR**	**AIBR**	RIVL	RIVL	**AIBL**	**AIBR**	**AIBR**	**AVER**
3	RIVR	**RIMR**	106	**AVEL**	**RIMR**	**AVEL**	**AIBR**	**AIBL**	RMEV	**RIML**
4	**AIBL**	RMED	RIVL	106	VB01	AVBR	RIGL	RIVL	VB02	RMEV
5	**RIML**	RIVR	ASKR	SIBDR	RMED	105	**AVEL**	SIBVL	128	VA01
6	DB02	DB02	**AIBL**	**RIML**	**AIBL**	**RIMR**	51	**AVEL**	15	**AVEL**
7	109	OLQVL	RID	**RIMR**	**AVER**	**AVER**	ASKL	SABD	127	127
8	URYDR	**AIBL**	ASKL	**AVER**	AVBR	RMEV	SABVR	RIVR	54	**AIBR**
9	OLQDL	OLQDL	**RIMR**	ASKR	**AIBR**	VB01	RIGR	OLQDL	ASKR	ASKR
10	73	VA01	104	RIVL	18	32	59	ALA	119	128

Table 4: Comparison of Top 10 rankings for **Reverse-1** behavior

	Worm 1		Worm 2		Worm 3		Worm 4		Worm 5	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	RMED	**AVEL**	**AIBR**	RIVR	**AIBR**	RIVL	**AIBR**	**RIML**	**RIMR**	**AVER**
2	**AIBR**	RMED	**AIBL**	**AVEL**	**AIBL**	**AIBR**	**AIBL**	**AIBR**	**AIBR**	**AIBR**
3	**RIMR**	RIVR	SIBDR	**AIBR**	89	**AVEL**	SIBVL	**AIBL**	VB02	**RIMR**
4	DB02	**RIMR**	**RIMR**	DB01	RMED	VB01	RMED	SIBVL	RMED	RMEV
5	**AIBL**	**AIBR**	VB03	RIVL	VB01	**AVER**	ASKL	**AVEL**	RMEV	**AVEL**
6	35	AVBR	RIVR	**RIMR**	16	**AIBL**	**RIML**	RMED	127	RMED
7	DB01	RIVL	DB02	**RIML**	**RIML**	RMED	37	RIVL	50	VA01
8	38	OLQDL	ASKL	VB03	**AVER**	105	SABD	ASKL	54	VB02
9	109	35	DB01	SIBDR	RIVL	**RIML**	SAAVR	AVBR	VA01	AVBL
10	VB01	VB01	RID	40	120	**RIMR**	AVBR	SAAVR	120	87

Table 5: Comparison of Top 10 rankings for **Reverse-2** behavior

	Worm 1		Worm 2		Worm 3		Worm 4		Worm 5	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	**RIMR**	**RIMR**	**RIMR**	**RIMR**	**RIML**	**AIBR**	**AIBL**	**AIBR**	**RIML**	VA01
2	**RIML**	**RIML**	**AIBR**	**RIML**	**AVEL**	RMED	**AIBR**	RMED	VA01	**AIBR**
3	**AVEL**	**AVEL**	**RIML**	**AIBR**	**AVER**	**AIBL**	DA01	**AIBL**	**RIMR**	**RIML**
4	**AIBR**	RMED	**AVEL**	RIVR	RIVL	**AVER**	AS01	DA01	**AIBR**	RMED
5	**AVER**	DB01	RIVR	**AVER**	126	**RIML**	ASKL	**RIMR**	**AVEL**	RMEV
6	**AIBL**	VA01	**AIBL**	DB01	31	105	**RIMR**	AS01	**AVER**	AVBL
7	VA01	103	RIGL	**AVEL**	**AIBR**	AVBR	AVFR	AVBL	ALA	**AIBL**
8	OLQDL	**AIBR**	40	SIBDR	89	**RIMR**	28	OLQVL	126	**RIMR**
9	DB02	DB02	108	ASKR	118	RIVL	**AVEL**	RIVL	1	**AVER**
10	109	**AIBL**	ASKR	VB02	128	ASKR	OLQVL	RMER	128	**AVEL**

Table 6: Comparison of Top 10 rankings for **Sustained reversal** behavior

	Worm 1		Worm 2		Worm 3		Worm 4		Worm 5	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	RMED	RMED	SIBDR	RIMR	RMED	**AIBR**	RMED	**AIBR**	AVBL	RMEV
2	DB02	RIMR	RMED	RIML	105	RMED	AVFR	RMED	127	**AIBR**
3	AVBR	RMEV	DB01	**AIBR**	AVBR	**AIBL**	SIBVL	**AIBL**	RMEV	AVBL
4	RMEV	AVEL	**AIBL**	RIVR	RMEV	AVBR	RIS	DA01	87	VA01
5	VB01	RIML	RMEL	AVEL	67	105	AVBL	AS01	ASKR	RIML
6	DB01	**AIBL**	66	AVER	VB02	AVER	**AIBL**	RIMR	125	**AIBL**
7	RMER	DB02	VB03	VB01	VB01	RMER	RMEL	AVBL	VB02	RIMR
8	URYDL	AVBR	VB01	VB02	RIS	ASKR	VB01	RMER	ALA	AVEL
9	93	VA01	RIVR	RMED	RMER	83	RMER	SIBVL	120	RMED
10	RIVR	DB01	DB02	ASKL	82	67	**AIBR**	RIS	1	127

Table 7: Comparison of Top 10 rankings for **Slowing** behavior

	Worm 1		Worm 2		Worm 3		Worm 4		Worm 5	
Place	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**	**IG**	**SHAP**
1	**RIVL**	**RIVR**	SIBDR	**RIVR**	**RIVL**	**RIVL**	AIBR	**RIVL**	**RIVL**	**RIVL**
2	AVEL	**RIVL**	AIBR	**RIVL**	AIBR	RMED	**RIVL**	RIML	VA01	127
3	RIMR	AVEL	AVEL	RIML	AVER	AIBR	AIBL	ASKL	26	VA01
4	**RIVR**	RMEV	ASKL	SIBDR	AIBL	54	ASKL	RMED	RIMR	RMED
5	OLQDL	RMED	RIML	DB01	AVEL	AVER	RIML	AVEL	AIBR	RIMR
6	OLQVL	OLQVL	RIMR	40	RIML	29	AVBR	AVBR	127	26
7	AIBR	RIMR	AIBL	AIBR	54	VB02	28	AIBR	15	RMEV
8	AIBL	OLQVR	**RIVR**	OLQDL	29	AVEL	SABD	**RIVR**	54	AIBR
9	93	OLQDL	OLQDR	RIMR	ASKR	32	AVEL	SAAVL	ALA	AVER
10	RMEV	VA01	VB03	AVEL	RMEV	AIBL	37	28	28	RIML

Table 8: Comparison of Top 10 rankings for **Ventral turn** behavior

As demonstrated in the tables above, the reference neurons identified by Kato et al. (2015) [10] frequently emerge within the highest-ranking positions across multiple behavioral states and individual worms. However, in several instances, key reference neurons fail to reach the top ranks. Furthermore, the feature importance profiles exhibit noticeable variability both between individual worms and across different feature attribution methods—specifically, SHAP [16] and Integrated Gradients [33].

Moreover, the results indicate that neurons idefntifies as crucial for one specific behavior also frequently appear as highly important in others, even when they were not explicitly highlighted by Kato et al. for that particular movement. This outcome demonstrates that neurons do not operate in an isolated manner; rather, they are integrated into a shared population dynamics framework and perform coordinated roles in driving state transitions [37]. Consequently, neuron families such as RMED, AVE, RIM, or AVB consistently recur across all behavioral rankings.

Additionally, the feature rankings reveal not only known reference neurons but also unidentified neurons (e.g., neurons 127 and 61 in Forward locomotion), which occasionally occupy the top rank. This indicates that their underlying neural activity provides a crucial predictive signal for the model during these behavioral states.

The recurrent presence of these biological ground-truth neurons suggests that BunDLe-Net's latent representations successfully capture the true underlying neural dynamics of *C. elegans* locomotion, rather than relying on spurious correlations or noise. Nevertheless, as highlighted in interpretability literature [15], a mere visual inspection of top-10 attribution rankings is insufficient to establish definitive conclusions regarding model alignment and trustworthiness [18]. Because these qualitative results cannot fully validate the model's robustness, a rigorous quantitative correlation analysis was incorporated to evaluate the hypothesis.

As mentioned in the **Diminishing saliency of the neurons** sub-section, this type of data frequently encounter the challenge of tail noise dominance. Consistent with the low-dimensional manifold hypothesis in systems neuroscience [6] and feature pruning studies [25], only a compact subset of leading neurons (approximately 5–10%) drives the model's behavioral predictions, while the remaining majority contributes negligible signal or background noise. Under such heavy tail noise, conventional metrics like the global Spearman rank correlation yield distorted and untrustworthy results, as random rank permutations among near-zero attributions disproportionately dilute the correlation of genuine leading drivers [40]. To mitigate this issue and isolate the biologically meaningful signal, we incorporated two top-weighted similarity metrics established in information retrieval and interpretability literature: the Top-k Jaccard Similarity and the Rank-Biased Overlap (RBO) coefficient [40].

4.1.3 Correlation between worms

As stated by Mathis and Mathis (2026) [18], joint-modeling model should perform similar across different subjects so we are assured that it generalized the behaviour of the brain, To test that the inter-worm correlation was calculated. The results for 10 unique worm pairs were shown in the table 9 below. The 'Mean $\rho\pm$ SD' columns display the average correlation coefficient and standard deviation. The 'Sig. Pairs' columns indicate the number of pairs that achieved statistical significance ($p < 0.05$)

Behavioral State	Mean RBO ($p = 0.9$) ± SD	Mean Top-10 Jaccard ± SD
Global (All states)	0.397 ± 0.118	0.339 ± 0.091
Dorsal Turn	0.253 ± 0.115	0.211 ± 0.086
Forward	0.331 ± 0.110	0.240 ± 0.086
Reverse-1	0.364 ± 0.084	0.254 ± 0.074
Reverse-2	0.368 ± 0.069	0.304 ± 0.085
Sustained Reversal	0.415 ± 0.126	0.333 ± 0.104
Slowing	0.376 ± 0.093	0.259 ± 0.111
Ventral Turn	0.406 ± 0.083	0.260 ± 0.055
No state	0.131 ± 0.104	0.075 ± 0.078

Table 9: Summary of inter-worm attribution consistency across all defined behavioral states (**SHAP**)

Behavioral State	Mean RBO ($p = 0.9$) ± SD	Mean Top-10 Jaccard ± SD
Global (All states)	0.350 ± 0.106	0.240 ± 0.086
Dorsal Turn	0.234 ± 0.114	0.110 ± 0.079
Forward	0.174 ± 0.076	0.077 ± 0.038
Reverse-1	0.255 ± 0.071	0.154 ± 0.074
Reverse-2	0.395 ± 0.081	0.187 ± 0.069
Sustained Reversal	0.389 ± 0.142	0.265 ± 0.099
Slowing	0.301 ± 0.132	0.167 ± 0.114
Ventral Turn	0.380 ± 0.097	0.247 ± 0.080
No state	0.134 ± 0.064	0.108 ± 0.056

Table 10: Summary of inter-worm attribution consistency across all defined behavioral states (**Integrated gradients**)

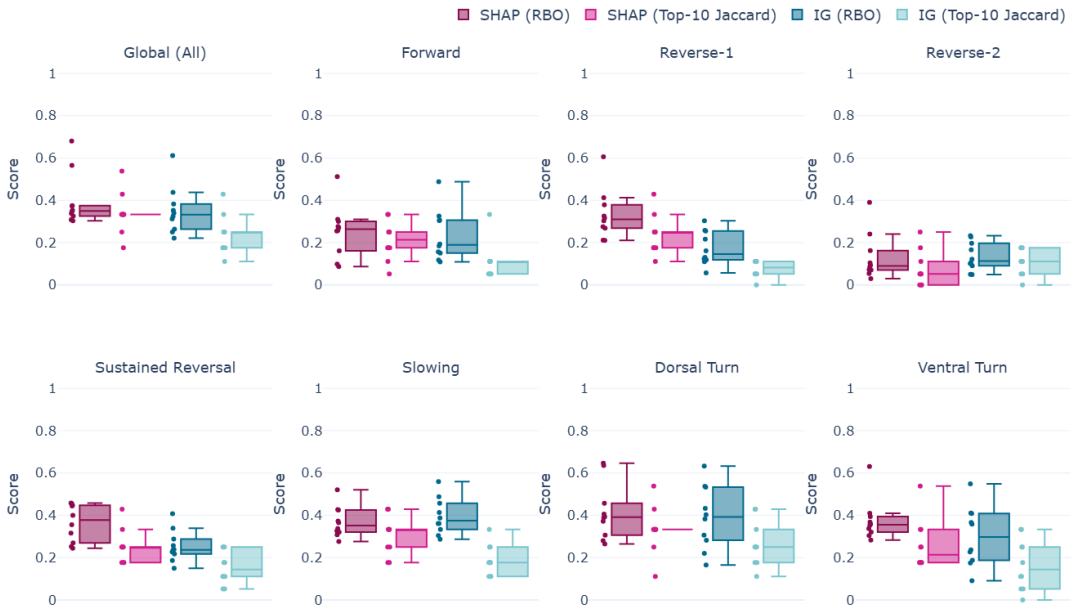


Figure 2: Comparassion of SHAP and IG results consistency between worms per each behaviour

At first glance, the results might seem unsatisfying, as the absolute consistency values across all comparisons are relatively low. However, in the context of biological data from *C. elegans*, we view these results as highly realistic and valuable.

Firstly, the coefficients across both XAI methods drop significantly for the "No state" category. During periods without a defined behavioral state, there should not be any specific command neurons taking the lead in the worm's brain [10]. This means that across different worms, we should not expect to see any recurring ranking patterns. This hypothesis perfectly aligns with our findings, the consistency for the "No state" category is virtually non-existent.

Compared to the "No state" baseline, defined behavioral states exhibit up to four times stronger consistency. This is a highly positive outcome, suggesting that BunDLe-Net's predictions are grounded in consistent biological trends rather than the idiosyncratic noise of individual worms brains. Moreover, results occurring mostly in the 0.30–0.40 range are well-supported by the concept of neural degeneracy [7]. A Top-10 Jaccard value in this range indicates that, on average, 3 to 4 neurons in the top ranking are shared across worms, while the rest vary. This clearly demonstrates that while executing the same behavior, different worms rely on the same 3 to 4 core command neurons, alongside different subsets of auxiliary neurons. This perfectly illustrates neural degeneracy, empirical evidence for which in *Caenorhabditis elegans* is reported by Trojanowski et al. (2014) [36].

Lastly, these results provide insight not only into BunDLe-Net's trustworthiness but also into the performance of the chosen XAI methods. To visualize the variance in inter-subject consistency and directly compare the explainer methods, we generated a head-to-head scatter plot array (Figure 3). Each data point represents the Rank-Biased Overlap for a single pair of worms. Points above the diagonal indicate SHAP is higher than IG.

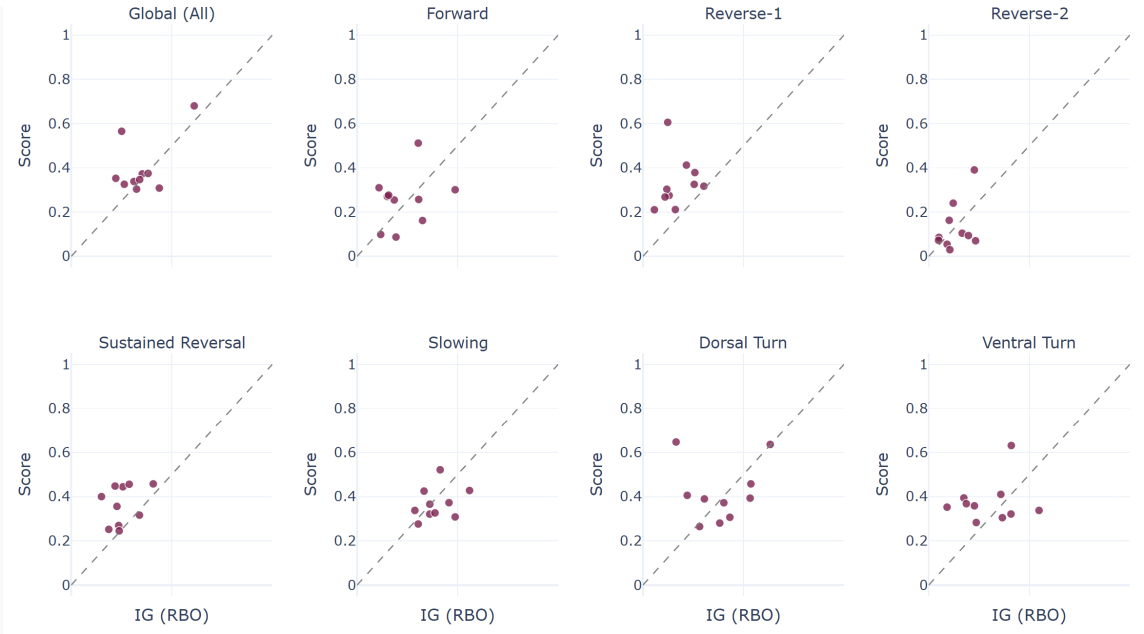


Figure 3: Compassion of SHAP and IG results consistency between worms per each behaviour

SHAP values have proven to be higher and significantly more stable than Integrated Gradients (IG) across almost all behavioral states. For example, during the Forward behavior, the IG Jaccard score drops to as low as 0.077, which is practically at the level of statistical noise. However, the scatter plot shows that its not case for all of the pairs of worms and the results are subjected to rather high variance in some categories like Dorsal Turn or Ventral turn. Nevertheless findings suggest that SHAP performs much better in filtering out biological noise to identify true predictive features.

4.1.4 Correlation between behaviours

In their whole-brain imaging study, Kato et al. (2015) [10] demonstrated that the primary mode of neural variance (PC1) separates forward and backward locomotion. Consequently, we hypothesize that feature attribution profiles between forward and backward states will exhibit lower correlations, whereas states belonging to the same functional family, such as Reverse-1, Reverse-2, and Sustained Reversal, will display strong positive correlations. Nevertheless, based on our prior observations, some degree of correlation is expected across behaviors, reflecting the continuous nature of low-dimensional neural manifolds.

Each cell in Table 11 and 12 displays Rank-Biased Overlap (RBO, $p = 0.9$) followed by the Top-10 Jaccard Similarity (J), averaged across all worms. Cell backgrounds are color-coded based on the RBO value to highlight relations: very weak (< 0.20, light orange), weak ($0.20 - 0.30$, light gray), moderate ($0.30 - 0.40$, light blue), strong ($0.40 - 0.50$, medium blue), and very strong (≥ 0.50, dark blue). The upper triangle is omitted for clarity.

Behavior	Fwd	Rev-1	Rev-2	Sust-Rev	Slow	Dorsal	Ventral
Fwd	1.00						
Rev-1	0.36 | J: 0.33	1.00					
Rev-2	0.31 | J: 0.23	0.29 | J: 0.21	1.00				
Sust-Rev	0.42 | J: 0.38	0.45 | J: 0.34	0.38 | J: 0.25	1.00			
Slow	0.37 | J: 0.31	0.48 | J: 0.43	0.35 | J: 0.22	0.61 | J: 0.47	1.00		
Dorsal	0.37 | J: 0.38	0.56 | J: 0.38	0.33 | J: 0.24	0.52 | J: 0.48	0.50 | J: 0.47	1.00	
Ventral	0.38 | J: 0.32	0.49 | J: 0.39	0.38 | J: 0.21	0.49 | J: 0.38	0.49 | J: 0.34	0.77 | J: 0.65	1.00

Table 11: Inter-behavior consistency matrix using **SHAP**.

Behavior	Fwd	Rev-1	Rev-2	Sust-Rev	Slow	Dorsal	Ventral
Fwd	1.00						
Rev-1	0.25 | J: 0.14	1.00					
Rev-2	0.33 | J: 0.21	0.22 | J: 0.11	1.00				
Sust-Rev	0.35 | J: 0.29	0.27 | J: 0.22	0.32 | J: 0.29	1.00			
Slow	0.33 | J: 0.18	0.32 | J: 0.29	0.36 | J: 0.28	0.43 | J: 0.36	1.00		
Dorsal	0.37 | J: 0.32	0.15 | J: 0.09	0.24 | J: 0.18	0.43 | J: 0.29	0.37 | J: 0.25	1.00	
Ventral	0.23 | J: 0.10	0.48 | J: 0.35	0.22 | J: 0.11	0.22 | J: 0.17	0.35 | J: 0.26	0.13 | J: 0.09	1.00

Table 12: Inter-behavior consistency matrix using **Integrated Gradients**.

The inter-behavior consistency is generally higher than initially anticipated. Notably, the SHAP attributions reveal a strong, shared neural substrate across nearly all behavioral states, whereas the Integrated Gradients (IG) results exhibit greater variance and overall lower overlap.

Interestingly, the *Forward* state maintains a moderate and uniform consistency with all other behaviors, including reversals. While one might intuitively expect opposing movements to utilize entirely distinct neural circuits, this shared overlap likely reflects the continuous, default nature of forward locomotion in *C. elegans*. It implies the existence of a baseline network state that other behaviors modulate, rather than completely replace. Similarly, within the family of reverse movements (*Reverse-1*, *Reverse-2*, *Sustained Reversal*), the internal consistency does not stand out as uniquely high compared to other cross-behavior pairs.

Instead, the highest feature overlaps in the SHAP evaluation are observed among *Slowing*, *Dorsal Turn*, and *Ventral Turn* movements. Most strikingly, there is a sharp divergence between the two explainer methods for specific behavioral pairs. Pairings that yield the highest consistency under SHAP, such as *Dorsal Turn* vs. *Ventral Turn* (RBO of 0.77), or *Reverse-1* vs. *Dorsal Turn*, simultaneously exhibit some of the lowest overlap scores when evaluated with IG. This stark contrast suggests that SHAP successfully identifies the shared underlying command structure of complex directional transitions, whereas IG is highly susceptible to biological noise and struggles to maintain attribution stability across structurally related states.

However, since those are only averaged results, valuable insight might come from also considering variance of the results. In the Figure 4, we decided to show boxplots of each pair of behaviours consistency for SHAP and IG vaules.

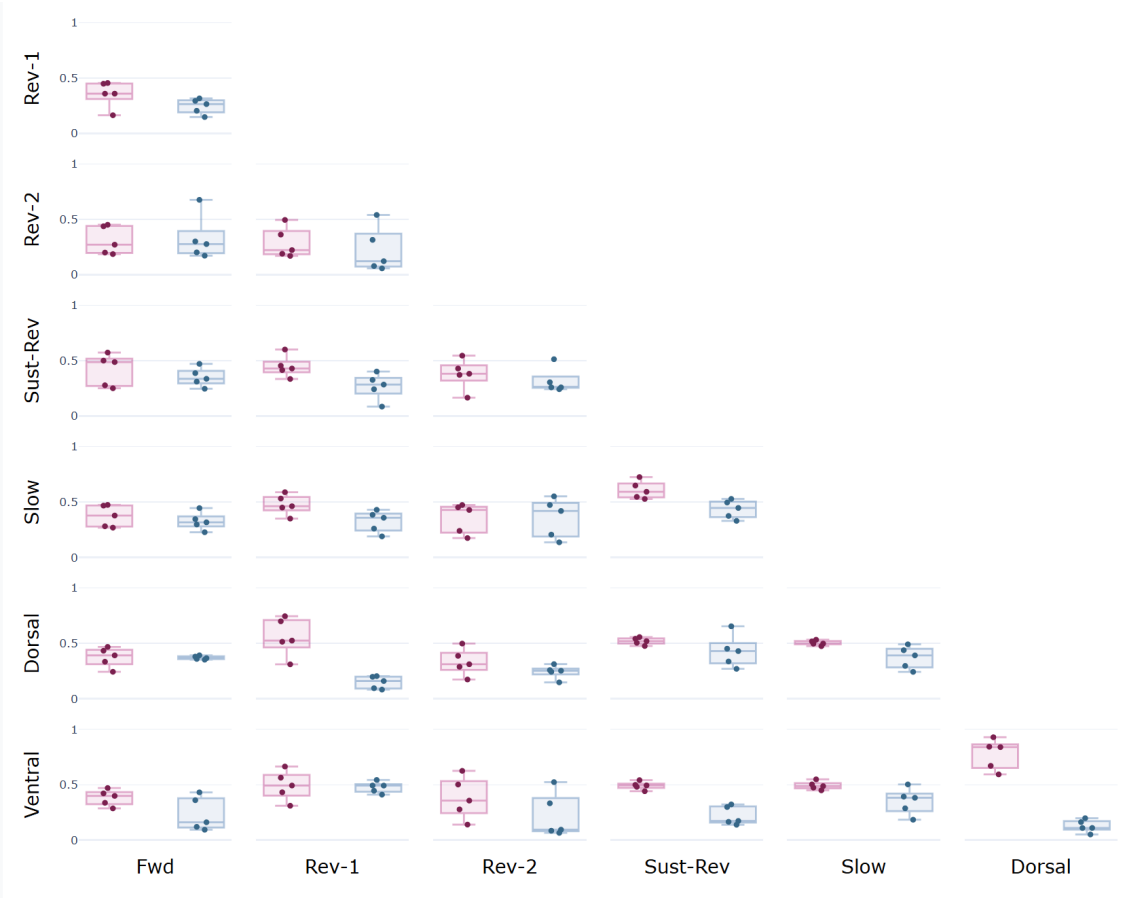


Figure 4: Compassion of SHAP and IG results consistency between behaviours per each worm

While the inter-subject variance appears broadly similar for both XAI methods across several simpler behavioral transitions, significant divergences emerge when examining complex movements, particularly those involving Dorsal and Ventral turns. Within these specific categories, the boxplot distributions exhibit striking differences in attribution stability. For instance, in comparisons such as *Dorsal Turn* vs. *Sustained Reversal* and *Ventral Turn* vs. *Slowing*, the SHAP values remain tightly clustered. This indicates a highly robust and consistent neural overlap across all individual subjects. In stark contrast, the Integrated Gradients (IG) scores for these same pairs display wide dispersion, underscoring the method's vulnerability to individual biological noise.

4.1.5 Summary

The main aim of this subsection was to assess whether BunDLe-Net is able to faithfully mirror the biological neural dynamics of *C. elegans*. We identified several 'ground truths' that should find confirmation across our results:

- Diminishing saliency of neuron importance
- High importance of core neurons (aligning with Kato et al., 2015) alongside neural degeneracy
- High consistency of feature importance across individual worms
- Different levels of consistency between specific pairs of behaviors

The results suggest that the model can mirror these biological principles well, though not perfectly. There are several deviations from the ground truths, for instance, in the unexpected consistency outcomes when comparing certain pairs of behaviors (e.g., the overlap between forward and reverse states). This suggests that while these results promisingly position BunDLe-Net as a trustworthy model, further and deeper research should be done in this area.

Furthermore, the findings are limited by the chosen explanation methods. As the authors of xCEBRA highlighted, this type of neural data is subject to extremely high correlation, which can heavily influence simple XAI methods [29]. Nevertheless, the performance across the conducted analyses clearly shows that SHAP significantly outperforms Integrated Gradients (IG), yielding higher and much more stable results.

4.2 When the most importnat neurons are working?

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